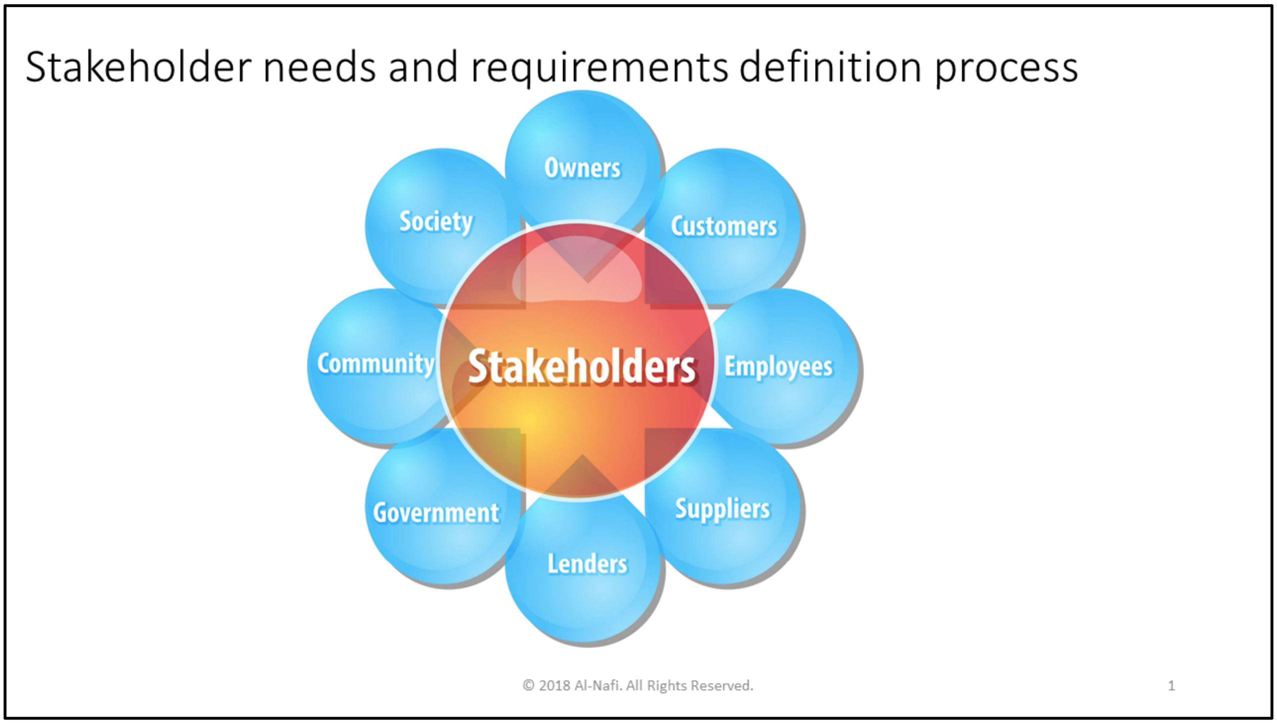
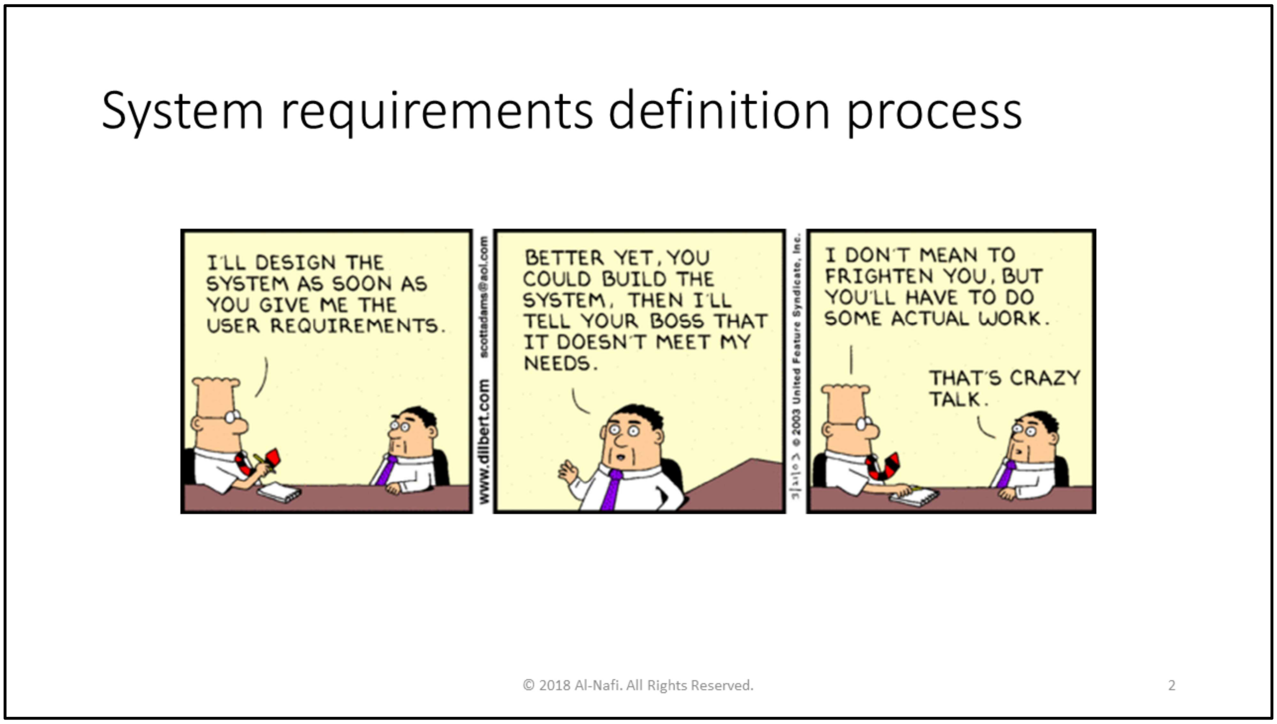


Stakeholder needs and requirements definition process

Owners
Society
Customers
Community
Stakeholders
Employees
Government
Suppliers
Lenders

Stakeholder needs and requirements definition process:
Defines the stakeholder security requirements that include the protection capability, security characteristics, and security-driven constraints for the system to securely provide the capabilities needed by users and other stakeholders.



System requirements definition process

I'LL DESIGN THE SYSTEM AS SOON AS YOU GIVE ME THE USER REQUIREMENTS.
BETTER YET, YOU COULD BUILD THE SYSTEM, THEN I'LL TELL YOUR BOSS THAT IT DOESN'T MEET MY NEEDS.
I DON'T MEAN TO FRIGHTEN YOU, BUT YOU'LL HAVE TO DO SOME ACTUAL WORK.
THAT'S CRAZY TALK.
www.dilbert.com
scottadams@aol.com
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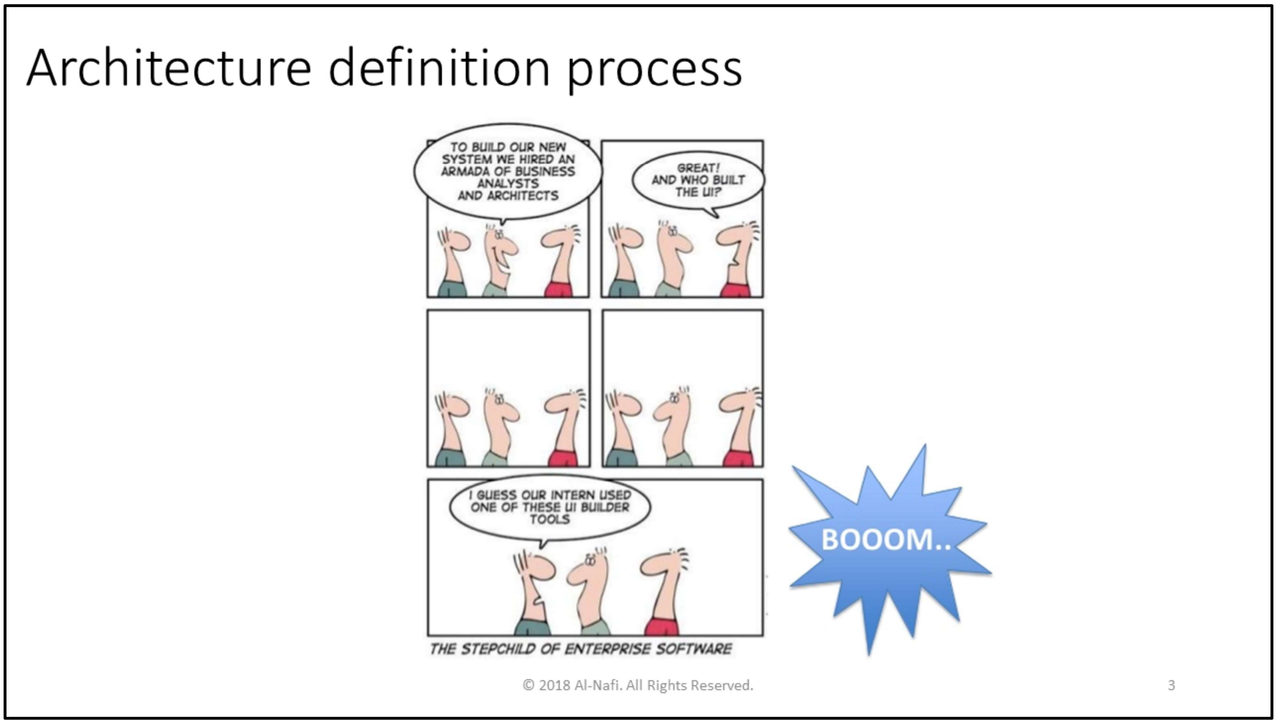
System requirements definition process:
Transforms the stakeholder security requirements into the system requirements that reflect a technical security view of the system.

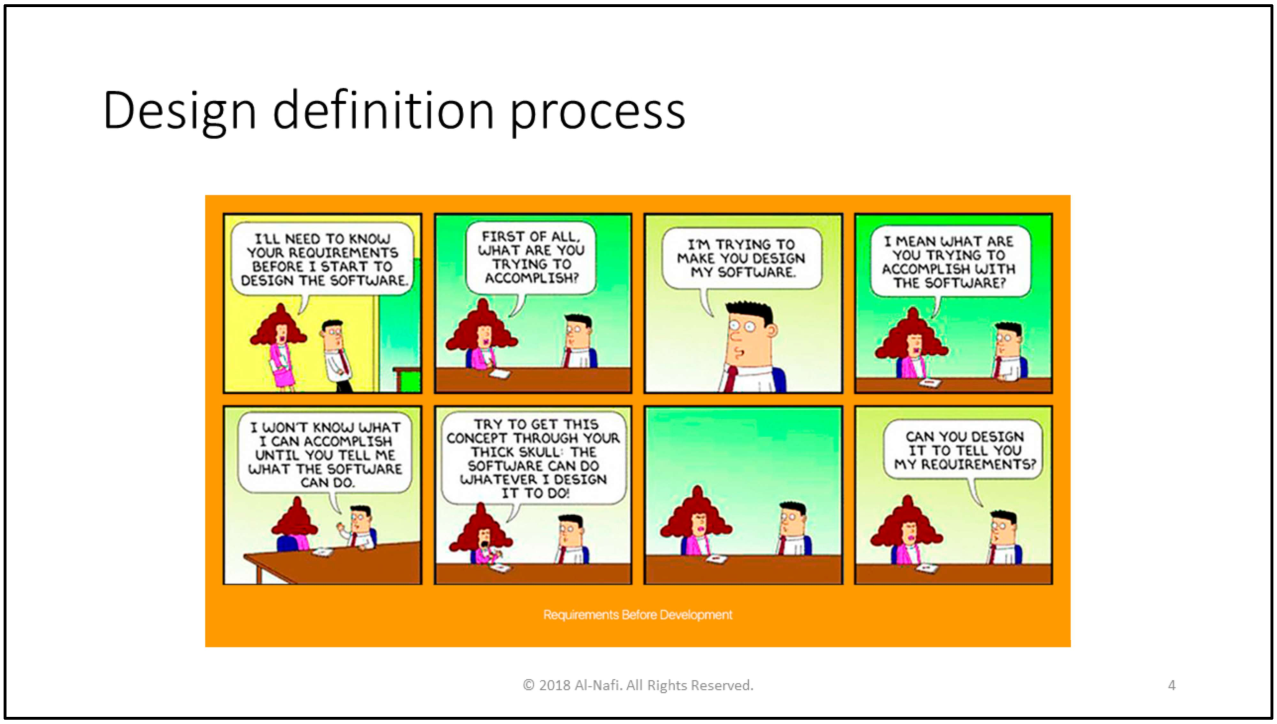


Architecture definition process

TO BUILD OUR NEW SYSTEM WE HIRED AN ARMADA OF BUSINESS ANALYSTS AND ARCHITECTS
GREAT! AND WHO BUILT THE UI?
I GUESS OUR INTERN USED ONE OF THESE UI BUILDER TOOLS
THE STEPCHILD OF ENTERPRISE SOFTWARE
BOOOM..

Architecture definition process:

Generates a set of representative security views of the system architecture alternatives to inform the selection of one or more alternatives.

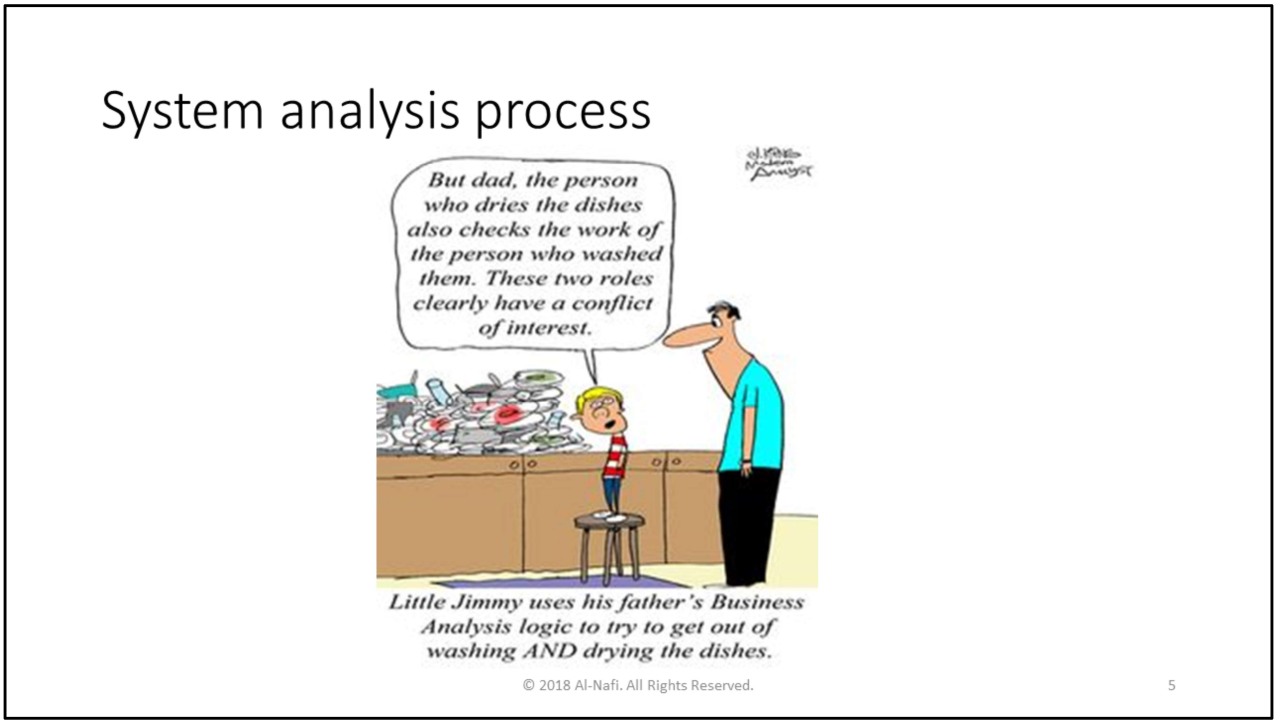


Design definition process

I'LL NEED TO KNOW YOUR REQUIREMENTS BEFORE I START TO DESIGN THE SOFTWARE.
FIRST OF ALL, WHAT ARE YOU TRYING TO ACCOMPLISH?
I'M TRYING TO MAKE YOU DESIGN MY SOFTWARE.
I MEAN WHAT ARE YOU TRYING TO ACCOMPLISH WITH THE SOFTWARE?
I WON'T KNOW WHAT I CAN ACCOMPLISH UNTIL YOU TELL ME WHAT THE SOFTWARE CAN DO.
TRY TO GET THIS CONCEPT THROUGH YOUR THICK SKULL: THE SOFTWARE CAN DO WHATEVER I DESIGN IT TO DO!
CAN YOU DESIGN IT TO TELL YOU MY REQUIREMENTS?

Requirements Before Development

Design definition process:
Provides security-related data and information about the system and its elements to enable implementation consistent with security architectural entities and constraints as defined in the models and views of the system architecture.

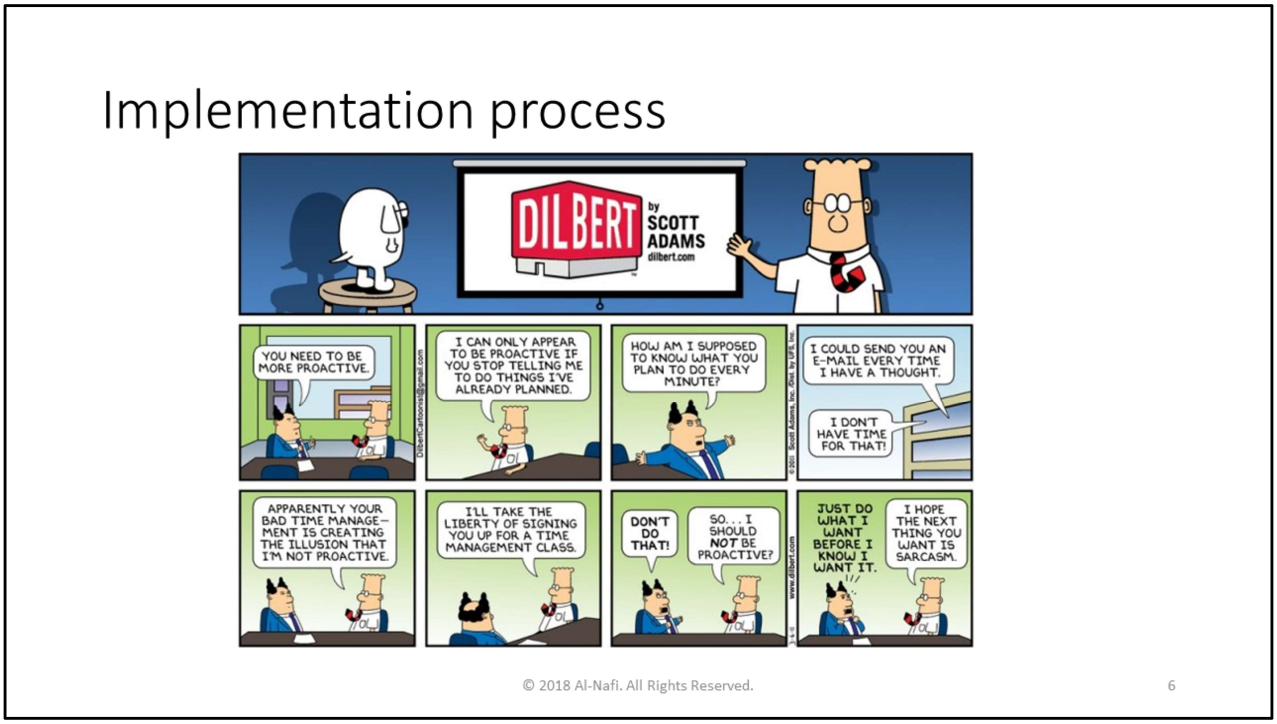


System analysis process

But dad, the person who dries the dishes also checks the work of the person who washed them. These two roles clearly have a conflict of interest.

Little Jimmy uses his father's Business Analysis logic to try to get out of washing AND drying the dishes.

System analysis process:
Provides a security view to system analyses and contributes specific system security analyses to provide essential data and information for the technical understanding of the security aspects of decision-making.

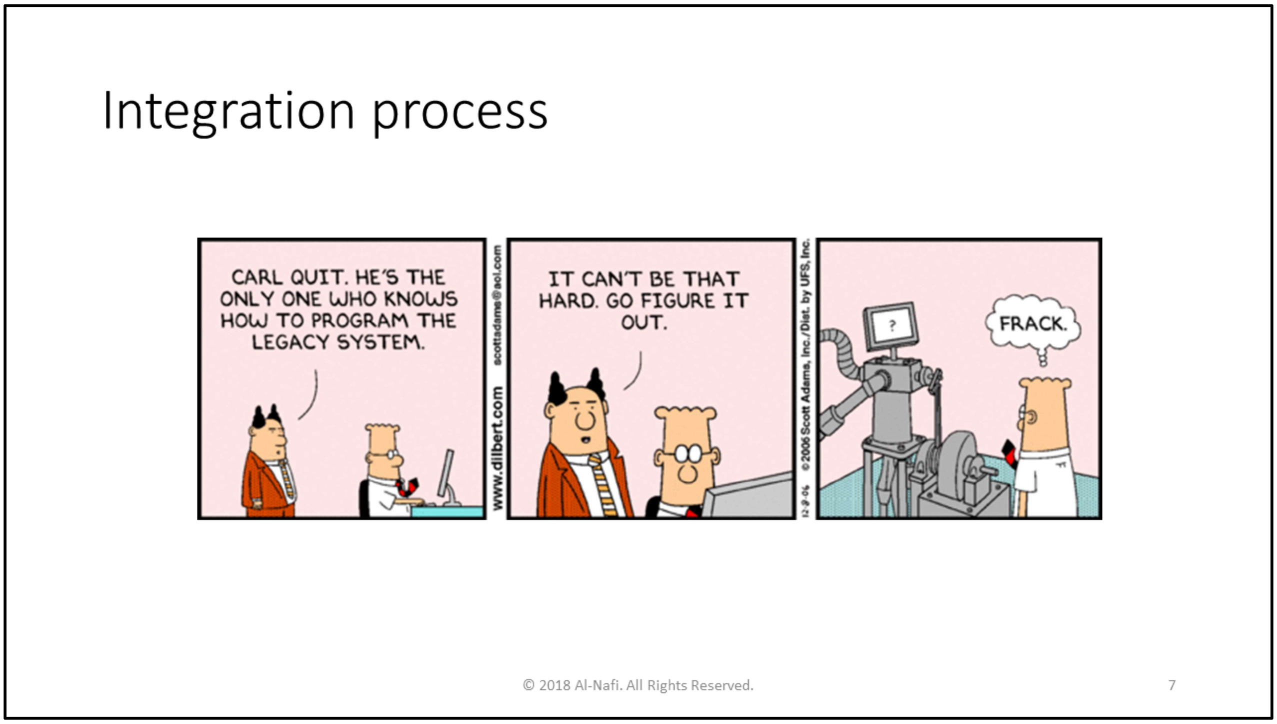


Implementation process

DILBERT
by SCOTT ADAMS
dilbert.com
YOU NEED TO BE MORE PROACTIVE.
I CAN ONLY APPEAR TO BE PROACTIVE IF YOU STOP TELLING ME TO DO THINGS I'VE ALREADY PLANNED.
HOW AM I SUPPOSED TO KNOW WHAT YOU PLAN TO DO EVERY MINUTE?
I COULD SEND YOU AN E-MAIL EVERY TIME I HAVE A THOUGHT.
I DON'T HAVE TIME FOR THAT!
APPARENTLY YOUR BAD TIME MANAGEMENT IS CREATING THE ILLUSION THAT I'M NOT PROACTIVE.
I'LL TAKE THE LIBERTY OF SIGNING YOU UP FOR A TIME MANAGEMENT CLASS.
DON'T DO THAT!
SO... I SHOULD *NOT* BE PROACTIVE?
JUST DO WHAT I WANT BEFORE I KNOW I WANT IT.
I HOPE THE NEXT THING YOU WANT IS SARCASM.

Implementation process:

Realizes (implements, builds) the security aspects of all system elements.

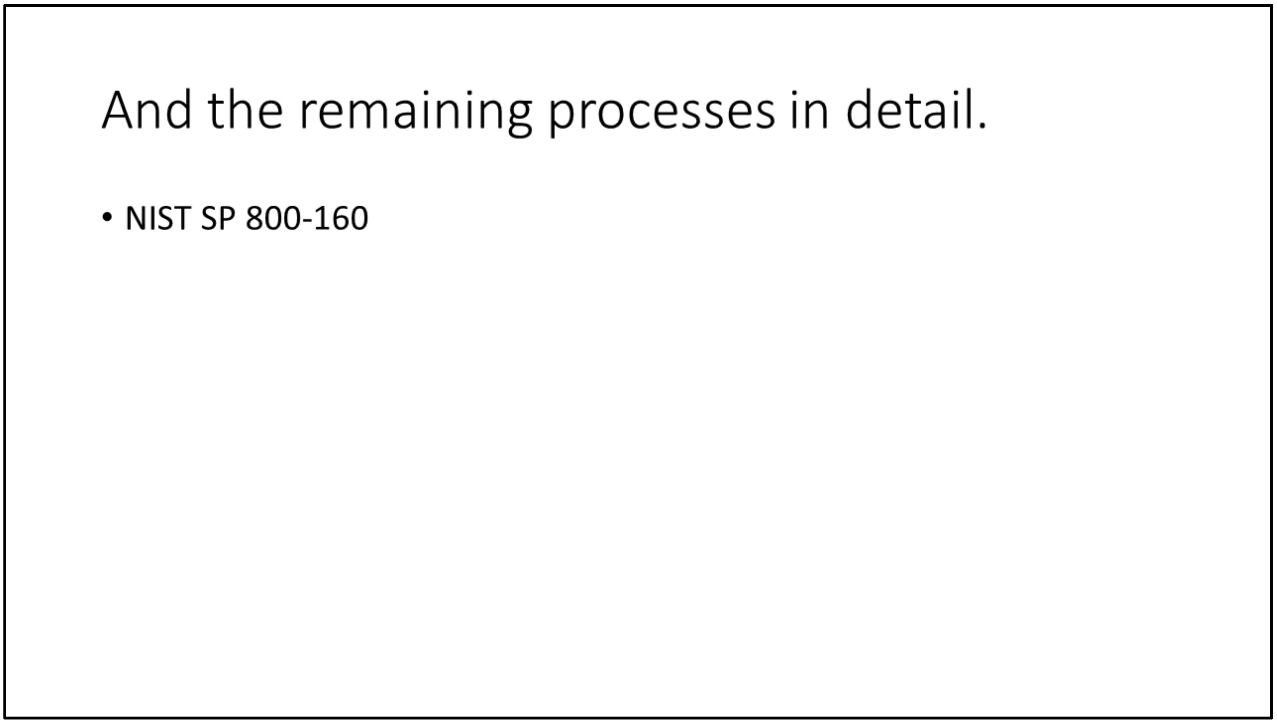


Integration process

CARL QUIT. HE'S THE ONLY ONE WHO KNOWS HOW TO PROGRAM THE LEGACY SYSTEM.
scottadams@aol.com
www.dilbert.com
IT CAN'T BE THAT HARD. GO FIGURE IT OUT.
© 2006 Scott Adams, Inc./Dist. by UFS, Inc.
12-9-06
?
FRACK.

Integration process:

Addresses the security aspects in the assembly of a set of system elements such that the realized system achieves the protection capability in a trustworthy manner as specified by the system security requirements and
in accordance with the system architecture and system design.



And the remaining processes in detail.

- NIST SP 800-160

Verification process:
Produces evidence sufficient to demonstrate that the system satisfies its security requirements and security characteristics with the level of assurance that applies to the system.

Validation process:
Provides evidence sufficient to demonstrate that the system, while in use, fulfills its business or mission objectives while being able to provide adequate protection of stakeholder and business or mission assets; minimize or contain asset loss and associated consequences; and achieve its intended use in its intended operational environment with the desired level of trustworthiness.

Transition process:
Establishes a capability to preserve the system security characteristics during all aspects of an orderly and planned transition of the system into operational status.

Operation process:

Establishes the requirements and constraints to enable the secure operation of the system in a manner consistent with its intended uses, in its intended operational environment, and for all system modes of operation.

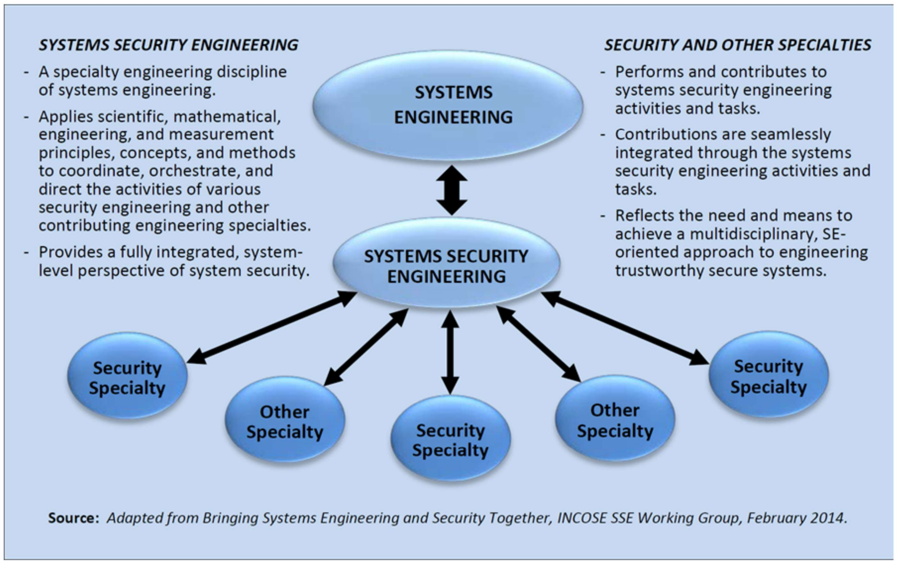
Maintenance process:

Establishes the requirements and constraints to enable maintenance elements to sustain delivery of the specified system security services and provides engineering support to maintenance elements.

Disposal process:

Provides for the security aspects of ending the existence of a system element or system for a specified intended use. It accounts for the methods and techniques used to securely handle, transport, package, store, or destroy retired elements to include the data and information associated with the system or contained in system elements.

System Engineering and Other Specialty Engineering Disciplines



SYSTEMS SECURITY ENGINEERING

- A specialty engineering discipline of systems engineering.
- Applies scientific, mathematical, engineering, and measurement principles, concepts, and methods to coordinate, orchestrate, and direct the activities of various security engineering and other contributing engineering specialties.
- Provides a fully integrated, system-level perspective of system security.

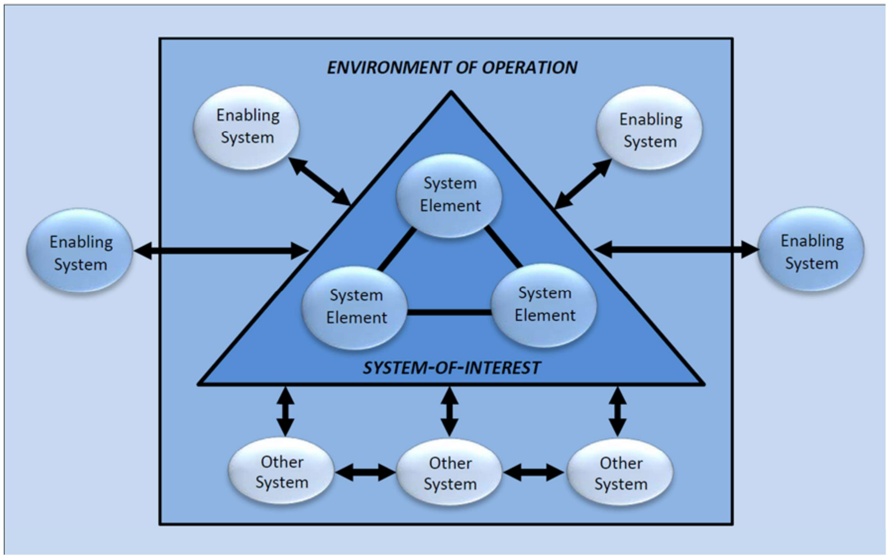
SECURITY AND OTHER SPECIALTIES

- Performs and contributes to systems security engineering activities and tasks.
- Contributions are seamlessly integrated through the systems security engineering activities and tasks.
- Reflects the need and means to achieve a multidisciplinary, SE-oriented approach to engineering trustworthy secure systems.

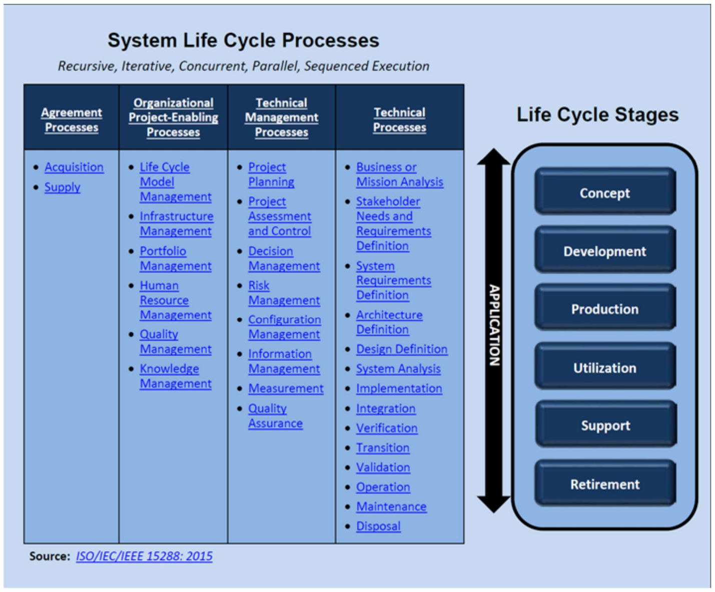
SYSTEMS ENGINEERING
SYSTEMS SECURITY ENGINEERING
Security Specialty
Other Specialty
Security Specialty
Other Specialty
Security Specialty

Source: *Adapted from Bringing Systems Engineering and Security Together, INCOSE SSE Working Group, February 2014.*

Security Engineering Environment of Operations


ENVIRONMENT OF OPERATION
Enabling System
Enabling System
Enabling System
Enabling System
System Element
System Element
System Element
SYSTEM-OF-INTEREST
Other System
Other System
Other System

System Life Cycle Processes



System Life Cycle Processes

Recursive, Iterative, Concurrent, Parallel, Sequenced Execution

Agreement Processes	Organizational Project-Enabling Processes	Technical Management Processes	Technical Processes
• Acquisition • Supply	• Life Cycle Model Management • Infrastructure Management • Portfolio Management • Human Resource Management • Quality Management • Knowledge Management	• Project Planning • Project Assessment and Control • Decision Management • Risk Management • Configuration Management • Information Management • Measurement • Quality Assurance	• Business or Mission Analysis • Stakeholder Needs and Requirements Definition • System Requirements Definition • Architecture Definition • Design Definition • System Analysis • Implementation • Integration • Verification • Transition • Validation • Operation • Maintenance • Disposal

Life Cycle Stages

- Concept
- Development
- Production
- Utilization
- Support
- Retirement

Source: *ISO/IEC/IEEE 15288: 2015*